

GD32 çŽ`âçfâ@žçŽ`è`æ`Ž ## ç»@æ ‡ âœ` macOS ä,Šä,° `GD32F303RCT6`
â¼€â`æ`ç`æ`â`°ä,€â¥—ä,â¾`èµ— Keilâ€ä,â¿@æ`âŽŸâš< `GD32F303Rxt6 KIT`
ç»@â½•çš,,â»°ä»¶â¼€â`â€æ`æž,,â»°â€çfšâ½•â`EéâEè`çŽ`âçfâ€
, æœ€ç»`â»šç%œ`â`â`â»¥€šè¿‡ ST-Link çfšâ½•çš,,â»°ä»¶â¼€â`â¿â»¶â¼š - `firmware/build/
gd32f303rct6_firmware.elf` - `firmware/build/gd32f303rct6_firmware.bin` - `firmware/build/
gd32f303rct6_firmware.hex` - `firmware/build/gd32f303rct6_firmware.map` ## â@žçŽ`æ—
¹â¼`æ¿,èš`çŽ`âçfâ€‡ç` Homebrew + Arm GCC + CMake + Ninja + ST-Link` â@žçŽ`â€, æ•
`â½`æ€è`â¿,â¿,â¿š 1. â½¿ç` Homebrew â@%œ€... macOS â`ç`çš,,âµEâ...¥â¼`â`¥â...
·é“â€, 2. â½¿ç` `arm-none-eabi-gcc` â½œä,° Cortex-M4 è£,æœ°âœâ`%ç¼—è`â™`â€, 3.
â½¿ç` CMake ç@¿ç`†â`¥ç`æ`ç`â€ç¼—è`é€
%œ¿¹â€è“â€æŽŸè,,šæœ—â`Eä°šç%œ`ç`Ÿæ`â€, 4. â½¿ç` Ninja
â½œä,æž,,â»°â`Žç«`î¼Eæ`è«æž,,â»°é€Ÿâ¿â`Eâ`é¿`â¿æ€šâ€, 5. â½¿ç` `stlink`
â`¥â...â,çš,, `st-info` â`E `st-flash` è`†â`â€çfšâ½•â`Eæ ¿âEâ¼€â`æ`â¿, 6.
âŽŸâŽ, `GD32F303Rxt6 KIT` â`â½œä,â`â`» SDK â¾`èµ—î¼Eä,âœ`â...¶ä,-
â¿@æ`¹â»â½•æ—â»¶â¿, ## â`¥â...é“â@%œ€... æœ—æœ°é€šè¿‡ Homebrew â@%œ€...
â»¥â¿,â`¥â...î¼š ``sh brew install cmake ninja arm-none-eabi-gcc stlink open-ocd ``
â¿,â`¥â...â½œç`î¼š - `cmake`î¼šç`Ÿæ`â¿â»¶â¿ç`æž,,â»°ç³»ç`Ÿâ€, -
`ninja`î¼šæ%œ¿è¿Eâ¿žé™...ç¼—è`â`Eé“â€æŽŸâ€, - `arm-none-eabi-gcc`î¼šç¼—è` Cortex-
M4 è£,æœ° C/ASM â»£ç`â€, - `arm-none-eabi-objcopy`î¼šâ»Ž ELF ç`Ÿæ`â¿.bin` â`E
`.hex`â€, - `arm-none-eabi-size`î¼šç`Ÿè¿¿ Flash â`E RAM â`ç`â€, - `st-info`î¼šæŽŸçæµ ST-
Link â`Eç»@æ ‡èš`ç%œ¿â¿¿â€, - `st-flash`î¼šé€šè¿‡ ST-Link â¿™â...¥â€è`â¿—
â`Eæ ¿âE Flashâ€, - `openocd`î¼šéç,,ç™è`fè`â`¥â...
·î¼Eæœ—æ—¿çfšâ½•æµç`â¿,»è¿â½¿ç` `stlink`â€, ## â`¥â...é“âEè`â@%œ€...
â¿ŽéâEè`â`½â»¿î¼š ``sh arm-none-eabi-gcc --version cmake --version ninja --version st-info
--version openocd --version `` ST-Link æŽŸçæµâ`½â»¿î¼š ``sh st-info --probe `` â¿žé™...
æŽŸçæµç»“æžœ¿¼š ``text Found 1 stlink programmers version: V2J37S7 serial:
XXXXXXXXXXXX flash: 262144 (pagesize: 2048) sram: 65536 chipid: 0x414 dev-type:
Flxx_HD `` è¿™è`æ~Ž ST-Link â`èç«` macOS æ£â¿,è`†â`â€æ¿,æ³»æ,,â¿¼šè¿™è¿Eçš,, `sram:
65536` æ~ STM32F1 â...¼â@¹â¿¿è@æšŸâ`Šâ€¼¼¿æœ—æ¿âŽŸâŽ, Keil â`¥ç`â¿æ`Ž
`IRAM(0x20000000,0x0C000)`î¼Eâ¿žé™...é“â€æŽŸè,,šæœ—æE%œ 48 KiB SRAM è...
¿ç¼¿â€, ## â`¥ç`ç»@â½•è@¾¿è¿ æ—â¿çžâ`¥ç`ç»@â½•â½¿â¿Ž¿¼š ``text .../firmware ``
ç»@â½•ç»“æž,,î¼š ``text firmware/ â”œâ”Eâ”E CMakeLists.txt â”œâ”Eâ”E cmake/ â”, â”œâ”Eâ”E
arm-none-eabi-gcc.cmake â”œâ”Eâ”E include/ â”, â”œâ”Eâ”E board.h â”, â”œâ”Eâ”E
gd32f30x_libopt.h â”, â”œâ”Eâ”E stdbool.h â”, â”œâ”Eâ”E stdint.h â”, â”œâ”Eâ”E systick.h
â”œâ”Eâ”E ld/ â”, â”œâ”Eâ”E GD32F303RCT6.ld â”œâ”Eâ”E src/ â”, â”œâ”Eâ”E board.c
â”, â”œâ”Eâ”E gd32f30x_it.c â”, â”œâ”Eâ”E main.c â”, â”œâ”Eâ”E systick.c â”œâ”Eâ”E startup/
â”œâ”Eâ”E startup_gd32f30x_hd_gcc.S `` â¿,éƒ`â`†èEè`£¿¼š -
`CMakeLists.txt`î¼šâ¿šâ¿%â¿â»¶â¿ç»@æ ‡â€çSDK è`â¾,,â€ç¼—
è`â¿,æ°â€è“â€æŽŸâ¿,æ°â`Eä°šç%œ`ç`Ÿæ`â€èš,,â™`â€, - `cmake/arm-none-eabi-
gcc.cmake`î¼šâ¿šâ¿%â¿âœâ¿%ç¼—è`â™`â`Eè£,æœ°æž,,â»°çŽ`âçfâ€, - `ld/
GD32F303RCT6.ld`î¼šâ¿šâ¿% Flashâ€RAMâ€â¿—âš`â...¥â¿£â`Eæµâ¿,fâ±€â€, -
`startup/startup\_gd32f30x\_hd\_gcc.S`î¼šGNU æ±¿ç¼—æ ¼â¼â¿â¿—âš`æ—â¿â»¶â¿Eä,æ—
â¿`é¿`è¿â€, - `include/gd32f30x\_libopt.h`î¼šé€%œâ¿æœ—â`¥ç`â¿žé™...â½¿ç`çš,, GD32
æ ‡â¿†â¿—è@¾âœ`â¿æ—â¿â¿, - `src/main.c`î¼šéâEè`â¿â»¶â¿,â¾âçŽ`â€, - `src/
board.c`î¼šæ¿ç¿ LED â`E USART0 â¿â¿â¿â¿â€, - `src/systick.c`î¼š1 ms
ç³»ç»Ÿèš,æ¿â¿Eâ»¶â¿—¶â¿½æ°â€, - `src/gd32f30x\_it.c`î¼šä,æ—æœ¿â¿¿â¿½æ°â€, ##
âŽŸâŽ, SDK çš,,â½¿ç`æ—¹â¼`âŽŸâŽ, SDK è`â¾,,î¼š ``text GD32F303Rxt6 KIT/3. SDK/
GD32F303RCT6 Demo/Library/GD32F30x_Firmware_Library_V2.1.0/Firmware ``
æœ—â`¥ç`â¿â¼â¿ç`â...¶ä,çš,,æ°ç`â¿Eâœ`æ—â¿â¿¶â¿¼Eä,â¿¿@æ`¹âŽŸç»@â½•â€
, â¼â¿`â¿...â¿¹âE...æ¿î¼š - CMSIS âœ`æ—â¿â¿¶â¿¼š Firmware/CMSIS` - GD32 è@¾â¿â¿æ—
â¿â¿¶â¿¼š Firmware/CMSIS/GD/GD32F30x/Include` - ç³»ç»ŸèŸâ¿â¿â¿â¿—
æ°ç`ç¿¼š`system_gd32f30x.c` - æ ‡â¿†â¿—è@¾âœ`âœ`æ—â¿â¿¶â¿¼š`Firmware/
GD32F30x_standard_peripheral/Include` - æ ‡â¿†â¿—

[illegible]

gd32f303rct6_firmware_readback.bin 0x08000000 3004 ``
æ”è¼fæœ-âœ°â°ä»¶â’Cè-»âžâ°ä»¶i¼š ``sh cmp firmware/build/
gd32f303rct6_firmware.bin firmware/build/gd32f303rct6_firmware_readback.bin `` `cmp` æ
— è¾¼“â†°i¼Cej”ç°ä,ä,ªæ-†ä»¶â@Câ...”ä,€è†’â€, SHA256 æ jéªCei¼š ``text
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware.bin
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware_readback.bin `` ä,ä,ª SHA256 â@Câ...”ä,€è†’i¼Cè-’æ~Žçfšâ½•â†...
â@¹â♦-â>žæ°-â€♦â♦-éªCè-♦â€, ## â½¼“â%º♦éªCè-♦ç»“è@º â·²ç»♦â@Cæ^♦çš,,éªCè-♦i¼š
- macOS â·¥â...·é“¾â@%ºèf...æ^♦âŠYâ€, - `arm-none-eabi-gcc` â♦-â»¥â@Cæ^♦ GD32
âºâ»¶ç¼-è-’â€, - CMake/Ninja â·¥ç<â♦-â»¥ç³â@šç”Yæ^♦ `elf/.bin/.hex/.map`â€, - ST-Link
â♦-â»¥èç«è-†â^«â€, - ç>@æ †èš-ç%º† Flash/SRAM ä¼æ♦-â♦-è-»â♦-â€, -
âºâ»¶â♦-â»¥æ^♦âŠYâ†™â...¥ `0x08000000`â€, - â†™â...¥â♦Ž `st-flash`
è†ªâŠ”æ jéªCæ^♦âŠYâ€, - æ%º<âŠ”è-»â>žæ-†ä»¶â,ŽâŽYâš<`.bin` â@Câ...”ä,€è†’â€
, éœ€è|çŽ°âœ°èš,â·Yçš,,éªCè-♦i¼š - RGB LED æ~â♦|æC%ºé|ºâ♦é—ªçf♦â€, - USART0
æ~â♦|è¾¼“â†°â♦-âŠ”ä¼æ♦-â’Cè `tick`â€, ## â...³é”@â♦-è^♦-ä,♦ä½¼ç”” Keil¼Cé♦¼â...♦
Windows/MDK çŽ-âçfä¾¼♦èµ-â€, - ä,♦ä¼@æ”¹`GD32F303Rxt6
KIT`i¼Cä¼♦æC♦âŽYâš<èµ,,æ-™â@Cæ·â€, - æ-ºâºº`firmware/
â½œä,ºç<ç««â·¥ç<`i¼Cä¾¼äºŽç%º^æœ-ç@jç♦†â’Cè¼ççš»â€, - ä½¼ç”” GCC
è†ªæœ%ºâ♦-âŠ”æ-†ä»¶â»¶â’Cé“¾æŽYè,,šæœ-i¼Cé♦¼â...♦â¾¼èµ- Keil/IAR ä,“ç””â♦-âŠ”æ-
†ä»¶â»¶â€, - ä½¼ç””æœ€âº♦æ †â††âºæ-†ä»¶â»¶â»¶è¼† Homebrew Arm GCC ç¼¼ºâº newlib âºæ-
†ä»¶â»¶çš,,é—@éç~â€, - ä¼ç·™ SWDi¼Cé™♦â½Žâºâ»¶è¼çèjCâ♦Žæ— æ³·é†♦æ-
ºçfšâ½•çš,,éšŽé™©â€, ## â♦Žç»âººè@@ 1. â¼,æžœœœèè|♦ä,²â♦æ—¥â¼—i¼CæŽYâ...¥
USB-TTL âºº`PA9/PA10` â¼¶â½¼ç”” `115200 8N1` æY¥çœ<è¾¼“â†°â€, 2.
â¼,æžœœœèè|♦æº♦ç ♦çºšè°fè-·i¼Câ♦-âYºâºŽ `openocd` âçžâš è°fè-·é...♦ç½@â€, 3.
â¼,æžœâ♦Žç»â¼·â...¥æ’âºšâº-è@¾¼i¼Câººâœ`gd32f30x\_libopt.h` ä,æC%ºœœ€âŠ â...
¥â¹âººâº-è@¾¼âºæ-†ä»¶â»¶i¼Câ¼¶âœ`CMakeLists.txt` ä,âŠ â...¥â¹âºº`.c` æº♦æ-†ä»¶â»¶â€, 4.
â¼,æžœœœèè|♦é†♦♦âºšçfšâ½•i¼Câ♦-ä¼ç·™`.hex` æ^-`.bin`
â½œä,ºâ♦-â,fäºšç%º©i¼Câ¼¶è@ºâ½• SHA256â€,